Lecture 13

- Page Rank
- Perron's Th.
- Power Method

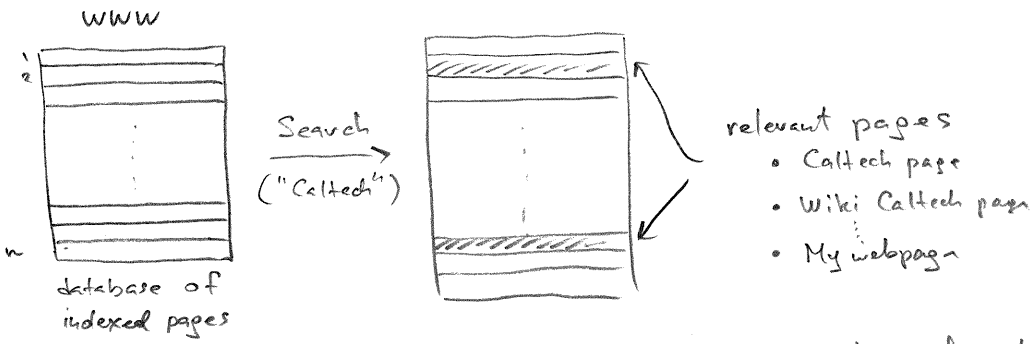
Page Rank

→ example of the eigenvalue problem in important real world application

→ application of Perron's Th

→ motivation to discuss the power method.

Problem Formulation:



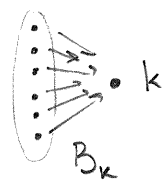
Q: In what order to show the relevant pages?

Idea: Let's rank ALL pages and show most important pages first.

$$\left\| \begin{array}{l} \text{Page } k \longmapsto x_k \text{ (importance score)} \\ i \text{ is more important than } j \iff x_i > x_j \end{array} \right\|$$

Q: How to assign scors x_k?

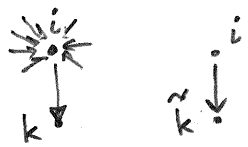
Idea: x_k is derived from k's "backlinks"



$\{i \mid i \to k\}$

- Attempt 0: $x_k = |B_k| = \sum_{i \in B_k} 1$ (0)

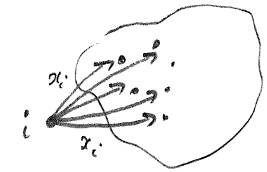
Drawback:



Intuitively, we want $x_k > x_{\tilde{k}}$, but according to (0), $x_k = x_{\tilde{k}}$

- Attempt 1: $x_k = \sum_{i \in B_k} x_i$ (1)

Drawback: a node i may become too influential by simply linking to many nodes.



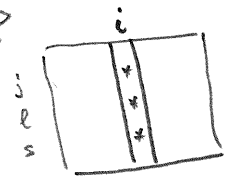
- Attempt 2: $\boxed{x_k = \sum_{i \in B_k} \frac{x_i}{n_i}}$ (2) $\left(= \sum_{i=1}^{n} L_{ki} x_i \right)$

Each node (page) has one vote, weighted by its score x_i, and evenly divided between all its links.

In matrix form: link matrix

(2') $\boxed{Lx = x}$, $L_{ki} = \begin{cases} \frac{1}{n_i} & i \in B_k \\ 0 & i \notin B_k \end{cases}$ $\quad i \to k$

The ranking problem reduced to the eigenvalue problem!



Q: Does (2') ≡ (2) have solution? (Is $\lambda = 1$ eigenvalue of L?)

$i \to j, l, s$ $\quad * = \frac{1}{n_i}$

$n_i = \#$links $\quad i \leftrightarrows n_i$

A: Yes (under certain condition)

Def A matrix $A \in \mathbb{M}_{n \times n}$ is column-stochastic if

- $A_{ij} \geq 0$
- $\sum_{i=1}^{n} A_{ij} = 1$, for all $j = 1, \dots, n$

Key observation: L is column stochastic (provided there is no "dangling" nodes)

Assume, there is no ← (without links) although not very important.

Th 1 If A is column stochastic $\Rightarrow$ $\lambda = 1$ is an eigenvalue

Proof $A^T \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \Rightarrow \lambda = 1$ is an eig.val of $A^T \Rightarrow$ of A.

Technical difficulties

① Nonuniqueness of x.

Let $V_1(L)$ be the eigenspace assoc. with $\lambda = 1$

For ranking, we want $\dim V_1(L) = 1$

- If a web is strongly connected $\Rightarrow \dim V_1(L) = 1$
  - can get from any page to any page

The case for WWW $\rightarrow$

- If a web has $r \geq 2$ connected component $\Rightarrow \dim V_1(L) \geq r$

$\Rightarrow$ not clear which vector from $V_1(L)$ to use for ranking

② Computing x.

Even if $\dim V_1(L) = 1$, how to find x?

L is $n \times n$, $n \sim 10^9$. $\rightsquigarrow$ power method.

Remedy for non uniqueness

Let's replace L with $\boxed{\tilde{L} = (1-\alpha) L + \alpha S}$, $\alpha \in [0,1]$

($\alpha = 0.15$ Google)

- $\alpha = 0 \Rightarrow \tilde{L} = L$
- $\alpha = 1 \Rightarrow \tilde{L} = S$, ultimately egalitarian case (all pages are equely imp.)

$$S = \begin{bmatrix} \frac{1}{n} & \cdots & \frac{1}{n} \\ \vdots & & \vdots \\ \frac{1}{n} & \cdots & \frac{1}{n} \end{bmatrix}$$

Properties:

1) $\tilde{L}$ is column stochastic

2) $\tilde{L}_{ij} > 0$, $\tilde{L}$ is positive matrix $\leftarrow$ key difference between L and $\tilde{L}$

Th 2 If A is column stochastic and positive $\Rightarrow \dim V_1(A) = 1$ (Perron's Th)

Let's study $V_1(A)$.

Lemma 1 If $v \in V_1(A) \setminus \{0\} \Rightarrow$ $v_i > 0 \ \forall i = 1..n$ or $v_i < 0 \ \forall i = 1..n$

Proof Assume v has component of diff. signs, and $A_{ij} > 0$

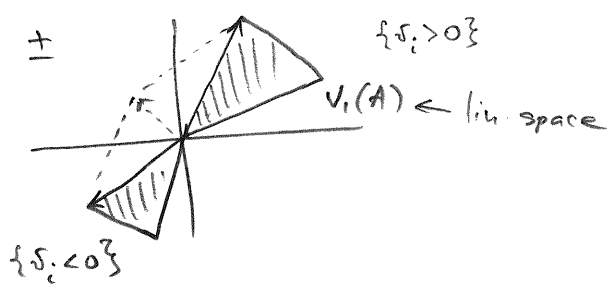
$v = Av \Rightarrow v_i = \sum_j A_{ij} v_j \Rightarrow$

$$|v_i| = \left| \sum_j A_{ij} v_j \right| < \sum_j |A_{ij} v_j| = \sum_j A_{ij} |v_j|$$

$$\sum_i |v_j| < \sum_i \sum_j A_{ij} |v_j| = \sum_j \underbrace{\sum_i A_{ij}}_{=1} |v_j| = \sum_j |v_j|$$

Contradiction!

- if $v_i \geq 0$ (all) $\Rightarrow v_i = \sum_j \overset{>0}{A_{ij}} \overset{\geq 0}{v_j} > 0$
- if $v_i \leq 0$ (all) $\Rightarrow v_i < 0$. $\square$



Lemma 2 If v_1 and v_2 are lin ind $\Rightarrow$ $\exists \alpha, \beta$: $u = \alpha v_1 + \beta v_2$ has components of diff. signs.

Proof: We want to find α, β: $\sum u_i = 0$

$$0 = \sum u_i = \alpha \underbrace{\sum (v_1)_i}_{d_1} + \beta \underbrace{\sum (v_2)_i}_{d_2}$$

- $d_1 = 0 \Rightarrow u = v_1$
- $d_1 \neq 0 \Rightarrow \beta = 1$, $\alpha = -d_2/d_1$ $\square$

Proof of Thm 2

Suppose that $\dim V_1(A) \geq 2$.
(since A is col-stoch, $V_1(A) \neq \{0\}$)

Then $\exists\, \delta_1, \delta_2 \in V_1(A)$ which are lin. indep.

Lemma 2: $\exists\, \alpha, \beta$ such that $\alpha\delta_1 + \beta\delta_2 \in V_1(A)$ has components of mixed signs. Contradiction with Lemma 1 $\Rightarrow$ $\dim V_1(A) = 1$ $\square$

So, we want to find $x \in V_1(\tilde{L})$:

(#) $\tilde{L}x = x$, $\tilde{L} = L(1-\alpha) + \alpha S$.

Since $\tilde{L}$ is col. stoch and positive, solution $\exists!$ (up to a multiplicative constant)

Q: How to solve (#)?

Power Method (aka Von Mises iteration)

Let $x_0 \in \mathbb{R}^n$ be any vector.

Let $\lambda_1 = 1, \lambda_2, \dots, \lambda_n$ be eigenvalues of $\tilde{L}$ and $\underset{x}{\delta_1} \dots \delta_n$ be an eigenbasis for $\tilde{L}$.

$\Rightarrow$ $x_0 = \alpha_1\delta_1 + \dots + \alpha_n\delta_n = \alpha_1 x + \sum_{i=2}^{n} \alpha_i \delta_i$

Consider the following iterative system

$$\begin{cases} x^{(0)} = x_0 \\ x^{(k)} = \tilde{L}\, x^{(k-1)} \end{cases}$$

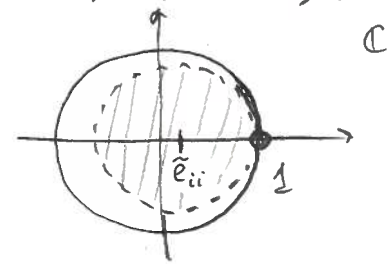
$$x^{(1)} = \tilde{L}\left(\alpha_1 x + \sum_{i=2}^{n} \alpha_i \delta_i\right) = \alpha_1 x + \sum_{i=2}^{n} \alpha_i \lambda_i \delta_i$$

In general, $x^{(k)} = \alpha_1 x + \sum_{i=2}^{n} \alpha_i \lambda_i^k \delta_i$

What can we say about $\operatorname{spec}(\tilde{L}) = \operatorname{spec}(\tilde{L}^T)$?

D_i: center $\tilde{\ell}_{ii}$
radius: $\sum_{j \neq i} \tilde{\ell}_{ij} = 1 - \tilde{\ell}_{ii}$

Gerschgorin disk for $\tilde{L}^T$ (row stochastic)



$\Rightarrow$ $|\lambda_i| < 1$ for $i \geq 2$

$\Rightarrow$ $x^{(k)} \longrightarrow \alpha_1 x$ as $k \to \infty$

Algorithm

1. Start from any x_0 (essentially we should not have $\alpha_1 = 0$ $\Leftrightarrow$ $x_0 \notin \operatorname{span}(\delta_2, \dots, \delta_n)$)
2. $x^{(k)} = \tilde{L}\, x^{(k-1)} \Big/ \| \tilde{L}\, x^{(k-1)} \|_\infty$

to avoid numerical issues

Iterate until convergence.
(until $\| x^{(k)} - x^{(k-1)} \|_\infty$ is small)

We can initialize the power method by

$(x_0)_i \sim U[0,1]$

i^{th} component

uniform distribution on $[0,1]$

In Matlab: $(x_0)_i$ = rand;
or
x_0 = rand(n,1);